PolSAR Feature Extraction Via Tensor Embedding Framework for Land Cover Classification

Bo Ren, *Member, IEEE*, Biao Hou, *Member, IEEE*, Jocelyn Chanussot, *Fellow, IEEE*, and Licheng Jiao, *Fellow, IEEE*

***Abstract*— Polarimetric synthetic aperture radar (PolSAR) as a typical multi-channel sensor can obtain refined geometrical and geophysical information. In the PolSAR land cover classification task, feature extraction is regarded as a critical step for the final classification. It can employ multi-modal features from the original scattering data, polarimetric target decomposition, and other transformation space. Then, how to efficiently combine multi-modal polarimetric information and extract discriminant features is an important challenge for PolSAR image processing. Graph embedding methods have become a significant technique to deal with feature extraction and dimensionality reduction (DR) problems in recent years. It provides a unified linearization framework in machine learning and other pattern recognition tasks. In this article, an extended tensor embedding framework is introduced to extract the intrinsic features for PolSAR land cover classification. First, each pixel is represented by a feature cube that is constructed by groups of polarimetric scattering signals and target decomposition features in a fixed size patch. Second, an intrinsic matrix is constructed to describe the original geometrical and statistical properties of the samples, and a penalty matrix is designed to represent some constraints. Third, the vector-based algorithms are transformed into tensor space in an unified framework and based on the pair of matrices to obtain the projection matrices in each mode by an iterative optimization process. The effectiveness of the proposed methods is demonstrated on three RADARSAT2 data sets covering the regions of Xi'an, San Francisco, and Flevoland, respectively. The visualization and quantification results show that the proposed method has superiority in land cover classification.**

***Index Terms*— Feature extraction, land cover classification, polarimetric synthetic aperture radar (PolSAR) image, tensor embedding framework.**

I. Introduction

POLARIMETRIC synthetic aperture radar (PolSAR) is a multiple channel radar system. It can acquire quad-polarization information to measure the targets in four receiving/transmitting models including HH, HV, VH, and VV and records more complete backscattering information than a traditional single-polarization synthetic aperture radar (SAR) [1], [2]. The obtained polarization information can describe the geometrical structure and the geophysical properties of targets. On the one hand, it provides more information for understanding and interpreting remote sensing images. With the development of imaging radar systems, more and more advanced airborne and spaceborne SAR systems have been launched, i.e., ALOS2, RADARSAT2, and TerraSAR. On the other hand, it also contributes to the advancement of PolSAR techniques and expands the application fields, which include military and civilian application domains, such as natural resource exploration, environment monitoring, agriculture control, forest vegetation change detection, and recognition and monitoring of military objects [3]–[6].

Land cover/use classification, as one of the most significant research directions in PolSAR data processing, has aroused many researchers' interest. For machine learning-based methods, feature extraction/fusion/selection and classifier design are two significant steps [7]. In this article, we focus on exploring the polarimetric properties and spatial information for achieving a group of distinctive features for land cover classification.

The most direct way to extract the polarimetric features is to utilize the backscattering coefficients or the parameters from the coherency/covariance matrix in land cover classification. In most image processing techniques, the nine parameters of the covariance/coherency matrix are always used as the input feature vector [8]. Some unsupervised and supervised methods, such as neural networks and fuzzy c-means, can initially be applied for PolSAR classification [9], [10]. Meanwhile, various target decompositions are proposed. The decomposition techniques are proposed to provide an interpretation of some physical constraints that generally reflect the average backscattering extracted from a sum of independent components [1]. Ferro-Famil *et al.* [11] initially combined H/A/α decomposition and Wishart classifier for the unsupervised classification of PolSAR data. Other terrain and land cover classification methods are also proposed utilizing the features from model-based decompositions [12]–[14], coherent decompositions [15], [16], and eigenvector-based decompositions [17]–[19]. The weights and coefficients of the components with specific scattering properties can all be regarded as features for land cover classifications.

A critical issue to improve the classification performance is the selection and extraction of the most representative and

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B. Ren, B. Hou, and L. Jiao are with the Key Laboratory of Intelligent Perception and Image Understanding of Ministry of Education of China, Xidian University, Xi'an 710071, China (e-mail: avcodec@hotmail.com).

J. Chanussot is with LJK, CNRS, Grenoble INP, Inria, Université Grenoble Alpes, 38000 Grenoble, France.

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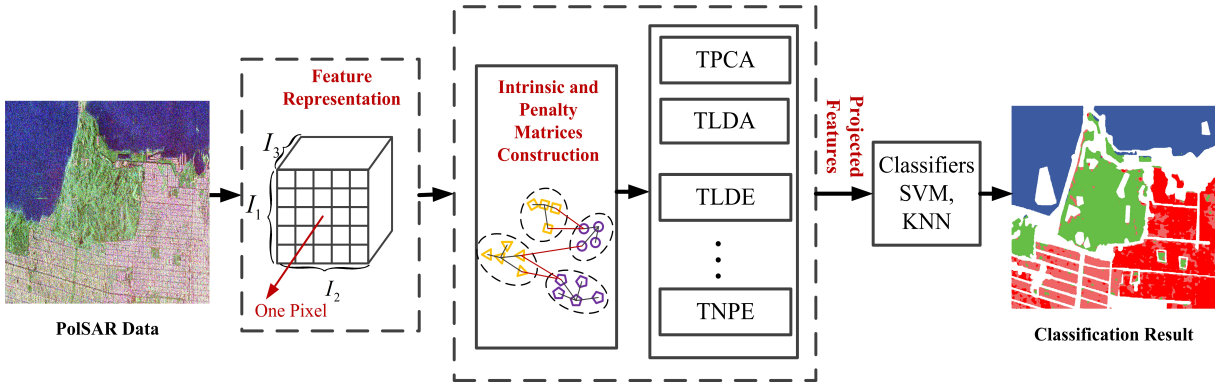

Fig. 1. Sketches of the proposed method.

discriminative features from the PolSAR data. Besides those polarimetric scattering- and target decomposition-derived original features, some methods also incorporate image processing-based features to capture the color, texture, and structural information from the PolSAR pseudo-color image [20]. The gray level co-occurrence matrix, Gabor wavelets, and other feature extraction techniques are employed to extract the spatial features from SAR images [21], [22]. They provide enough support for classifier learning and model construction. The classifiers for land cover classification include neural networks [23], sparse representation [24], and deep learning-based methods [25]. Those features not only provide more optional information for data interpretation, but also plague the designing of classifiers and reduce the algorithm efficiency.

How to take optimal advantage of those multimodal features is an important issue for PolSAR data interpretation. There is no conclusion regarding whether it is better to employ one view of features or multiview features in image processing techniques for PolSAR land cover classification. Those features from different views have been tested in the literature [20], and there are no absolutely better features from a specific extraction method for land cover classification. A convolutional neural network-based method tries to select a group of distinguished features from multiple polarimetric target decompositions as well, but it also shows that those selected features do not really belong to a fixed feature space [26]. Some researchers explore the intrinsic PolSAR data property and apply Laplacian Eigenmaps (LE) and a supervised graph embedding method to reduce the dimensionality [27], [28]. However, those methods all neglect the influence of the spatial information. In order to take into account the spatial relationship between the neighboring pixels, some researchers utilize tensor decomposition techniques in traditional linear dimension reduction (DR) algorithms to form the tensor-independent component analysis (TICA) and tensor local discriminant embedding (TLDE) methods [29], [30].

Nonlinear DR algorithms can explore the intrinsic manifold structures in data, including the algorithms of ISOMAP [31], locally linear embedding (LLE) [32], and LE [33]. Other machine learning algorithms are also developed recently, including the independent component analysis (ICA) [34], [35] and local tangent space alignment (LTSA) [36]. Those manifold-related algorithms all initially construct a local structure between pixels in manifold space and then utilize those local structures to embed pixels into a low-dimensional space. Those manifold algorithms can accurately learn the nonlinear structures of real data, but it has the drawback that the computation complexity is high and it is sensitive to outliers and noise. They are very difficult to apply in large-scale image classification and target recognition tasks.

Meanwhile, linear reduction algorithms are developed for obtaining projection matrices to solve the out-of-sample problem. Multidimensional scaling (MDS), neighbor preserving embedding (NPE) [37], locality preserving projection (LPP) [38], and linear LTSA (LLTSA) [39] are the linear versions of ISOMAP, LLE, LE, and LTSA, respectively. The principal component analysis (PCA) [40], linear discriminant analysis (LDA) [41], and local discriminant de-embedding (LDE) [42] are all linear algorithms employed to reduce the data dimensions under unsupervised and supervised conditions.

Tensor, as the high order of vector and matrix, can preserve the intrinsic data structure and offer a better explanation for the data processing. In particular, for PolSAR classification, the features of each pixel are always extracted from a patch around the pixel. Those features can be represented by a 3-D cube similar to other remote sensing image-processing methods [43]–[45], on which multilinear-algebraic techniques can then be applied. Recently, many DR techniques are modified into a high-order tensorial form to analyze the data structure. Those tensor-based DR algorithms include tensor LDA (TLDA) [46], [47], tensor LPP (TLPP) [48], TLDE [30], [48], tensor neighborhood preserving embedding (TNPE) [49], [50], tensor PCA (TPCA) [51], [52], and other modified versions of those tensorial DR algorithms under various constraints [30], [53].

Motivated by the multilinear representation and graph embedding frameworks [51], [54], [55], we assume that the aforementioned DR algorithms are just designed under different criteria, and can be unified within a tensorial reduction framework. Then, we propose a unified tensor embedding framework shown in Fig. 1 to extract the intrinsic features

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from multiple polarimetric data spaces under different supervised or unsupervised conditions. This framework is described by two matrices, which are intrinsic matrices computed by the similarity from the feature graph and the penalty matrix subject to different constraints. From other viewpoints, this also corresponds to the multi-linear version of the subspace learning framework, which can solve the out-of-sample problems in large tensor data processing. The polarimetric pixels can be represented by a tensor cube constructed from the neighbor pixels. First, it utilizes the similarity of pixels to construct the intrinsic matrix and the general constraints to build a penalty matrix. Second, the vector embedding techniques can be extended into a tensorial form in the proposed framework to process the features in high-order tensor space. Finally, the tensor embedding framework extracts the discriminative features from the redundant feature space for the final land cover classification task. It can be utilized to demonstrate the effectiveness of the multi-linear representation in polarimetric data processing under various statistics or geometry constraints.

This article is organized as follows. Section II introduces the tensor fundamental and a typical graph embedding framework and its linearization. Section III describes the proposed framework for PolSAR land classification. The experiments conducted on three data sets are presented in Section IV. Section V and VI present the conclusion and acknowledgment, respectively.

II. Preliminaries and Linear Embedding Framework

A. Tensor Fundamentals

A tensor is defined as a multidimensional array. Just as a vector is represented by a 1-D array of length n with respect to a given basis, any tensor with respect to a basis is represented by a multidimensional array. Some definitions are introduced as follows.

Definition 1: The number of dimensions (ways) of a tensor is its order, and each dimension is called a mode. Given a tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_M}$, the order of it is M. An element from it is $\mathcal{X}_{i_1,i_2,\ldots,i_M}$, and $1 \le i_n \le I_n$.

Definition 2: Unfolding: tensor to matrix transformation. A tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_M}$ can be transformed into a matrix by mode-n unfolding, the unfolded matrix is denoted $\mathbf{X}^{(n)} \in \mathbb{R}^{I_n (I_1 \times \cdots \times I_{n-1} \times I_{n+1} \times \cdots \times I_M)}$, where the column vectors of $\mathbf{X}^{(n)}$ are the mode-n vectors of $\mathcal{X}$.

Definition 3: Mode-n product: tensor matrix multiplication. The mode-n product of a tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_M}$ by matrix $\mathbf{U} \in \mathbb{R}^{P_n \times I_n}$ is equal to a tensor $\mathcal{Y} \in \mathbb{R}^{I_1 \times \cdots \times I_{n-1} \times P_n \times I_{n+1} \times \ldots \times I_M}$, and it is formulated as: $\mathcal{Y} = \mathcal{X} \times_n \mathbf{U}$. An element of $\mathcal{Y}$ is defined as: $\mathcal{Y}_{i_1 \times \cdots \times i_{n-1} \times p_n \times i_{n+1} \times \ldots i_M} = \sum_{i_n}^{I_n} \mathcal{X}_{i_1 \times i_2 \times \cdots \times i_M} \cdot \mathbf{U}_{p_n, i_n}$ The above mode-n product can also be rewritten by transforming into mode-n unfolding $\mathbf{Y}^{(n)} = \mathbf{U}\mathbf{X}^{(n)}$.

Definition 4: For a tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_M}$ and two matrices $\mathbf{U} \in \mathbb{R}^{P_n \times I_n}$ and $\mathbf{U} \in \mathbb{R}^{P_m \times I_m}$, the following equation holds $(\mathcal{X} \times_n \mathbf{U}) \times_m \mathbf{V} = (\mathcal{X} \times_m \mathbf{V}) \times_n \mathbf{U}$.

TABLE I
List of Symbols

Symbols	Description
x	Lowercase letters represent a scale.
$\mathbf{y}$	Boldface lowercase letters represent a vector.
$\mathbf{M}$	Boldface capital letters represent a matrix.
$\mathcal{X}$	Calligraphic letters represent a tensor.
$\times_n$	Tensor-matrix mode-n product operator.
$\langle \cdot, \cdot \rangle$	Inner product (the scalar product).
$\|\cdot\|_{\mathrm{F}}$	Frobenius norm .
$\mathbf{X}^{(n)}$	The mode-n unfolding matrix of tensor $\mathcal{X}$.
$\mathcal{X}_{i_1,i_2,\ldots,i_M}$	An element in tensor $\mathcal{X}$ with index $i_1, i_2, \ldots, i_M$.

Definition 5: The inner product of two same size tensors $\mathcal{X}, \mathcal{Y} \in \mathbf{R}^{I_1 \times I_2 \times \cdots \times I_M}$ is defined as $\langle \mathcal{X}, \mathcal{Y} \rangle = \sum_{i_1=1}^{I_1} \sum_{i_2=1}^{I_2} \cdots \sum_{i_M=1}^{I_M} \mathcal{X}_{i_1 \times i_2 \times \cdots \times i_M} \cdot \mathcal{Y}_{i_1 \times i_2 \times \cdots \times i_M}$.

Definition 6: The Frobenius norm of $\mathcal{X}$ is defined as $\|\mathcal{X}\|_{\mathrm{F}} = (\langle \mathcal{X}, \mathcal{X} \rangle)^{1/2}$. Then, we can get the distance between tensors $\mathcal{X}$ and $\mathcal{Y}$ as $\mathrm{dist}(\mathcal{X}, \mathcal{Y}) = \|\mathcal{X} - \mathcal{Y}\|_{\mathrm{F}}$.

Table I lists some symbols that will be employed throughout this article.

B. Graph Embedding Methods and Their Linearization

Given a set of data $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N] \in \mathbb{R}^{F \times N}$, where F is the number of features and N is the number of samples. The graph embedding methods seek low-dimensional features that are embedded in the original high-dimensional space [54], [56], [57]. The target is to find a function $f(\cdot)$ to transform the high-dimensional samples $\mathbf{X}$ into a low-dimensional space representation $\mathbf{Y} = [\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_N] \in \mathbb{R}^{P \times N}$, in which the vector y_i corresponds to the sample x_i and P denotes the number of low dimensions. This function can be nonlinear in manifold learning techniques and linear in some linear projection algorithms.

The main idea of graph embedding is to find an undirected graph $G = \{\mathbf{X}, W\}$ that can accurately characterize the similarity among the input samples under various constraints, which is also called the intrinsic graph. $\mathbf{X}$ is the vertex of the graph and $\mathbf{S} \in \mathbb{R}^{N \times N}$ is the similarity matrix, which is computed by pairwise vertexes. Then, graph embedding can obtain the low-dimensional representation based on the graph-preserving criterion. The objective function is as follows:

$$\begin{aligned}
\mathbf{Y}^* &= \underset{\mathrm{s.t.tr}(\mathbf{YBY})=c}{\arg\min} \sum_{i,j=1}^{N} \|\mathbf{y}_i - \mathbf{y}_j\|^2 S_{ij} \\
&= \underset{\mathrm{s.t.tr}(\mathbf{YBY}^{\mathrm{T}})=c}{\arg\min} \sum_{i=1}^{N} \mathbf{y}_i^T D_{ii} \mathbf{y}_i^T - \sum_{i,j=1}^{N} \mathbf{y}_i^T S_{ij} \mathbf{y}_i^T \\
&= \underset{\mathrm{s.t.tr}(\mathbf{YBY}^{\mathrm{T}})=c}{\arg\min} \mathrm{tr}(\mathbf{YDY}^{\mathrm{T}}) - \mathrm{tr}(\mathbf{YSY}^{\mathrm{T}}) \\
&= \underset{\mathrm{s.t.tr}(\mathbf{YBY}^{\mathrm{T}})=c}{\arg\min} \mathrm{tr}(\mathbf{YLY}^{\mathrm{T}}) \qquad (1)
\end{aligned}$$

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in which $\mathbf{D}$ is a diagonal matrix with each diagonal element denoted as $D_{ii} = \sum_{j=1}^{N} S_{ij}$ and the Laplacian matrix $\mathbf{L} = \mathbf{D} - \mathbf{S}$. This objective function ensures the similarity preservation property if the neighbor points $\mathbf{x}_i$ and $\mathbf{x}_j$ are relatively similar (a bigger value of S_{ij}), then their low-dimensional representation $\mathbf{y}_i$ and $\mathbf{y}_j$ will be fairly similar to. The Laplacian matrix represents the related manifold structure of data and is computed from the intrinsic graph. $\mathbf{B}$ is a constraint matrix that can be defined under the relationship between the samples from a penalty matrix G' or simply be a diagonal matrix for a scale normalization. When the constrain matrix $\mathbf{B}$ equals $\mathbf{D}$, it has the constraint $\mathrm{tr}(\mathbf{YDY}^{\mathrm{T}})$ in the objective function. It is the LE. Linearization means it needs to learn a transformation matrix $\mathbf{P} \in \mathbb{R}^{F \times P}$ that maps the high-dimensional data into the low-dimensional space as follows:

$$\mathbf{Y} = \mathbf{P}^{\mathrm{T}}\mathbf{X}. \tag{2}$$

Then, the objective function can be reformatted in linearization

$$\begin{aligned}\mathbf{P}^* &= \underset{\mathrm{s.t.tr}(\mathbf{P}^{\mathrm{T}}\mathbf{XBX}^{\mathrm{T}}\mathbf{P})=c}{\arg\min} \sum_{i,j=1}^{N} \left\|\mathbf{P}^{\mathrm{T}}x_i - \mathbf{P}^{\mathrm{T}}x_j\right\|^2 S_{ij} \\ &= \underset{\mathrm{s.t.tr}(\mathbf{P}^{\mathrm{T}}\mathbf{XBX}^{\mathrm{T}}\mathbf{P})=c}{\arg\min} \mathrm{tr}(\mathbf{P}^{\mathrm{T}}\mathbf{XLX}^{\mathrm{T}}\mathbf{P}).\end{aligned} \tag{3}$$

The linearization can well solve the out-of-sample problem. After obtaining the projection matrix, the new input high-dimensional samples can be directly mapped into the low-dimensional space without additional calculation of optimization of the objective function.

III. Proposed Algorithms for PolSAR Classification

A. Tensor Embedding Framework

PolSAR data, as in many other real-world applications, can be well represented by tensors. Considering the feature cube of pixels, a tensor representation, used instead of unfolding them into a high-dimensional vector, allows the effective representation of the spatial information of the data, overcoming the curse of dimensionality. To totally analyze the PolSAR data, it can extract features from different approaches in multimodality, such as the image processing methods for pseudoimages, the parameters of polarimetric target decomposition, and the original polarimetric scattering property.

Given a set of tensors $\{\mathcal{X}_i\}_{i=1}^{N} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_M}$, in which I_m is the dimension of the m-mode and M is the order of the tensor. The tensor embedding framework seeks to find M projection matrices $\mathbf{U}_k \in \mathbb{R}^{P_k \times I_k}, k = 1, 2, \ldots, M$ for transforming the dimension in each mode. Then, it obtains the tensor $\{\mathcal{Z}_i\}_{i=1}^{N} \in \mathbb{R}^{P_1 \times P_2 \times \cdots \times P_M}$. The tensor can be projected by the following equation:

$$\mathcal{Z}_i = \mathcal{X}_i \times_1 \mathbf{U}_1 \times_2 \mathbf{U}_2 \times_3 \cdots \times_M \mathbf{U}_M. \tag{4}$$

The projection matrices map the data from the original dimension and keep the intrinsic topological structure in the low dimension. Then, we can extend the linear embedding methods into a high-order DR framework, (1) can be reformulated with a tensor form

$$\begin{aligned}&\mathbf{U}_k^*|_{k=1}^{M} \\ &= \arg\min F(\mathbf{U}_1, \mathbf{U}_2, \ldots, \mathbf{U}_M) \\ &= \arg\min \sum_{i,j=1}^{N} \|(\mathcal{X}_i - \mathcal{X}_j) \times_1 \mathbf{U}_1 \times_2 \mathbf{U}_2 \times \cdots \times_M \mathbf{U}_M\|_F^2 S_{ij} \\ &\quad \mathrm{s.t.} \sum_{i=1}^{N} \|\mathcal{X}_i \times_1 \mathbf{U}_1 \times_2 \mathbf{U}_2 \times \cdots \times_M \mathbf{U}_M\|_F^2 B_{ij} = c.\end{aligned} \tag{5}$$

For optimizing the objective function, one needs to change the high-order programing problem into a matrix form, i.e., according to the alternative iteration of each transform matrix $\mathbf{U}_f$ and mode-f flattening of this high-order tensor into a matrix form. Then, one tensor sample can be projected in all modes except fth

$$\mathcal{Y}_i = \mathcal{X}_i \times_1 \mathbf{U}_1 \times \cdots \times_{f-1} \mathbf{U}_{f-1} \times_{f+1} \mathbf{U}_{f+1} \cdots \times_M \mathbf{U}_M. \tag{6}$$

The above objective function can be converted to

$$\begin{aligned}\mathbf{U}_k^*|_{k=1}^{M} &= \arg\min F(\mathbf{U}_1, \mathbf{U}_2, \ldots, \mathbf{U}_M) \\ &= \arg\min \sum_{i,j=1}^{N} \|\mathcal{Y}_i \times_f \mathbf{U}_f - \mathcal{Y}_i \times_f \mathbf{U}_f\|_F^2 S_{ij} \\ &\quad \mathrm{s.t.} \sum_{i=1}^{N} \|\mathcal{Y}_i \times_f \mathbf{U}_f\|_F^2 B_{ij} = c.\end{aligned} \tag{7}$$

It will be transformed by mode-f unfolding $\mathcal{Y} \times_f \mathbf{U}_f \Leftrightarrow \mathbf{U}_f \mathbf{Y}^{(f)}$, and get the matrix form. Then, (7) can be reformulated as

$$\begin{aligned}\mathbf{U}_k^*|_{k=1}^{M} &= \arg\min F(\mathbf{U}_1, \mathbf{U}_2, \ldots, \mathbf{U}_M) \\ &= \arg\min \sum_{i,j=1}^{N} \left\|\mathbf{U}_f \mathbf{Y}_i^{(f)} - \mathbf{U}_f \mathbf{Y}_j^{(f)}\right\|_F^2 S_{ij} \\ &\quad \mathrm{s.t.} \sum_{i=1}^{N} \left\|\mathbf{U}_f \mathbf{Y}_i^{(f)}\right\|_F^2 B_{ij} = c.\end{aligned} \tag{8}$$

We can obtain the trace form of function as follows:

$$\begin{aligned}&\mathbf{U}_f^*|_{f=1}^{M} \\ &= \arg\min F(\mathbf{U}_1, \mathbf{U}_2, \ldots, \mathbf{U}_M) \\ &= \arg\min \mathrm{tr}\left(\mathbf{U}_f \sum_{i,j=1}^{N} \left(\left(\mathbf{Y}_i^{(f)} - \mathbf{Y}_j^{(f)}\right)\left(\mathbf{Y}_i^{(f)} - \mathbf{Y}_j^{(f)}\right)^{\mathrm{T}} S_{ij}\right)\mathbf{U}_f^{\mathrm{T}}\right) \\ &\quad \mathrm{s.t.tr}\left(\mathbf{U}_f \sum_{i,j=1}^{N} \left(\mathbf{Y}_i^{(f)}\left(\mathbf{Y}_i^{(f)}\right)^{\mathrm{T}} B_{ij}\right)\mathbf{U}_f^{\mathrm{T}}\right) = c.\end{aligned} \tag{9}$$

After obtaining this reformulated function, we can utilize a pair of matrices $\mathbf{H}_1$ and $\mathbf{H}_2$ to unify the objective function into a unified framework. The first matrix $\mathbf{H}_1$ can be computed from the intrinsic graph and some statistical property. It is also called the intrinsic matrix. The second matrix $\mathbf{H}_2$ is constructed under the constraint of statistical or geometrical properties, so is defined as the penalty matrix. In (9), we have $\mathbf{H}_1 = \sum_{i,j=1}^{N} ((\mathbf{Y}_i^{(f)} - \mathbf{Y}_j^{(f)})(\mathbf{Y}_i^{(f)} - \mathbf{Y}_j^{(f)})^{\mathrm{T}} S_{ij})$ and

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$\mathbf{H}_2 = \sum_{i,j=1}^{N} (\mathbf{Y}_i^{(f)}(\mathbf{Y}_i^{(f)})^{\mathrm{T}} B_{ij})$. When $B_{ij} = D_{ii} = \sum_{j=1}^{N} S_{ij}$, Equation 9 defines the objective function of TLPP. The construction of those two matrices is not only limited to a fixed unsupervised construction method, but can also be formed under the supervised constraint and data distribution or other prior information.

B. Optimization of the Proposed Method

After defining two matrices $\mathbf{H}_1$ and $\mathbf{H}_2$, the objective function obtains a unified model

$$\mathbf{U}_k^*|_{k=1}^M = \arg\min \mathrm{tr}\left(\mathbf{U}_f \mathbf{H}_1 \mathbf{U}_f^{\mathrm{T}}\right) \quad \text{s.t.tr}\left(\mathbf{U}_f \mathbf{H}_2 \mathbf{U}_f^{\mathrm{T}}\right) = c. \tag{10}$$

It shows that this function has no closed-form optimization solution. Then, it needs to initialize each projection matrix $\mathbf{U}_k|_{k=1}^M$ and iteratively optimize them for obtaining the optimal solution. In general, it can be solved directly by generalized eigenvalue decomposition $\mathbf{H}_1\alpha = \lambda \mathbf{H}_2\alpha$, and can get the P_f eigenvectors associated with the P_f non-zero smallest eigenvalues from the eigenvalue decomposition as the optimal solution. It can also be reformulated into a ratio model and according to achieving the maximum value of max $\mathrm{tr}((\mathbf{U}_f \mathbf{H}_2 \mathbf{U}_f^{\mathrm{T}})(\mathbf{U}_f \mathbf{H}_1 \mathbf{U}_f^{\mathrm{T}})^{-1})$ for obtaining the optimal result. But $\mathbf{U}_f$ corresponds to the eigenvectors associated with the largest P_f eigenvalues. Then, it can be approximately solved by transforming into the eigenvalue decomposition problem.

However, in real applications, the inverse matrix does not always guarantee convergence in optimization. When optimizing the generalized eigenvalue decomposition and searching for the smallest P_f non-zero eigenvalues λ_{P_f}, it always produces eigenvalues close to zero. It weakens the convergence in iteration of each projection matrix $\mathbf{U}_f$ and makes it hard to obtain an optimal solution for the final DR algorithm. Then, it can also be reformulated into a subtractive mode between the trace terms to add the constraint and construct a maximum objective function

$$\mathbf{U}_f^*|_{f=1}^M = \underset{\mathbf{U}_f}{\mathrm{argmax}}\, \mathrm{tr}\left(\mathbf{U}_f(\mathbf{H}_2 - \gamma \mathbf{H}_1)\mathbf{U}_f^{\mathrm{T}}\right) \tag{11}$$

where γ is a balance parameter to tune the influence of the intrinsic matrix on the whole objective function. It clearly has a direct solution via the eigenvalue decomposition problem

$$\mathbf{D}\alpha = \lambda\alpha \tag{12}$$

in which $\mathbf{D} = \mathbf{H}_2 - \gamma \mathbf{H}_1$. In different tensorial DRs, we can tune the parameter γ and construct the corresponding matrix $\mathbf{H}_1$ and $\mathbf{H}_2$ under different statistical properties and supervised information. With this tensorial framework, we can extract intrinsic PolSAR features. This framework is summarized in Algorithm 1.

Algorithm 1 Overall Optimization of the Tensor Embedding Framework

Input: Randomly select three-order tensor samples $\mathcal{X}_i \in \mathbb{R}^{I_1 \times I_2 \times I_3}, i = 1, 2, \ldots, M$, reduced dimensions P_1, P_2, P_3, the number of neighbors k to construct the graphs (for TLDE, $k_1 = k_1 = k$), the label $l_1, l_2, \ldots, l_n \in \{1, 2, \ldots, C\}$ of each samples (for supervised methods), the parameter γ, the stop condition ε, and the maximum number of iterations $T_{\max}$.

Initialization: $\mathbf{U}_1, \mathbf{U}_2, \mathbf{U}_3$ are set as the identity matrix.

–TLPP, TNPE, TLDE: Construct the neighborhood graph G (G and G' for TLDE) and the corresponding similarity matrix S (S and S' for TLDE).

–TLDA: Calculate the overall mean of samples and the cth class mean.

For $t = 1$ to $T_{\max}$ **do**

  For $f = 1$ to 3 **do**

  Calculate tensor $\mathcal{Y}_i$ from Equation 6 and the corresponding mode-f unfolding matrix $\mathbf{Y}_i^f$.

  Update $\mathbf{H}_1$ and $\mathbf{H}_2$: TLPP, TPCA, TNPE, TLDE, and TLDA obtain the corresponding intrinsic and penalty matrices from Table II.

  Update $\{\mathbf{U}_f\}_{f=1}^3$ by solving the optimization problem of Equation 11.

  Update error $= \sum_{f=1}^3 \left\| \mathbf{U}_f^{k+1} - \mathbf{U}_f^k \right\|_2$

  End for

  Check iteration stop condition: if $error \le \varepsilon$, **break**

End for

Output: The optimal projection matrix $\{\mathbf{U}_f^*\}_{f=1}^3$.

C. Unifying Various Methods in Tensor Embedding Framework

Based on the unified framework, various linear DR techniques are extended to a tensorial version. As TLPP has been introduced as an example in Section II. In this section, we will introduce four tensorial algorithms, such as TPCA, TLPP, TLDE, and TLDA.

The algorithms mainly differ in how to construct the intrinsic and penalty graphs and further construct the corresponding matrices. Then, in the following section, we mainly introduce how to design a pair of matrices $\mathbf{H}_1$ and $\mathbf{H}_2$ and unify different typical tensor DR methods into the proposed framework.

1) TPCA: The tensorial PCA is also called the multilinear PCA. It is similar to the original PCA algorithm that constructs the projected tensor $\{\mathcal{Z}_i\}_{i=1}^N \in \mathbb{R}^{P_1 \times P_2 \times \cdots \times P_M}$ keeping the same variance as that in the original tensor space $\{\mathcal{X}_i\}_{i=1}^N \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_M}$. The variance is computed according to the total scatter, and we would like to obtain a set of mode-f matrices $\mathbf{U}_f|_{f=1}^M \in \mathbb{R}^{P_f \times I_f}$ to project the high-order tensor. The TPCA objective function can be formulated as a maximization problem to maximize the tensor scatter

$$\mathbf{U}_f^*|_{f=1}^M = \underset{\mathbf{U}_f}{\mathrm{argmax}} \sum_{i=1}^{N} \|\mathcal{Z}_i - \bar{\mathcal{Z}}\|_F^2. \tag{13}$$

The matrix $\bar{\mathcal{Z}} = (1/N)\sum_{i=1}^N \mathcal{Z}_i$ means the average of samples $\{\mathcal{Z}_i\}_{i=1}^N$. Because the samples should be preprocessed by the whitening approach, the mean of the projected matrices $\bar{\mathcal{Z}}$

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TABLE II
MATRIX $\mathbf{H_1}$ AND $\mathbf{H_2}$ IN TENSOR EMBEDDING FRAMEWORK

Matrix	TLPP	TNPE	TLDE	TLDA	TPCA
H1	$\sum_{i,j=1}^{N}\left(\left(\mathbf{Y}_i^{(f)}-\mathbf{Y}_j^{(f)}\right)\cdot\left(\mathbf{Y}_i^{(f)}-\mathbf{Y}_j^{(f)}\right)^{\mathrm{T}}(S_{ij})\right)$	$\sum_{i=1}^{N}\left(\left(\mathbf{Y}_i^{(f)}-\sum_{j=1}^{N}S_{ij}\mathbf{Y}_i^{(f)}\right)\cdot\left(\mathbf{Y}_i^{(f)}-\sum_{j=1}^{N}S_{ij}\mathbf{Y}_i^{(f)}\right)^{\mathrm{T}}\right)$	$\sum_{i,j=1}^{N}\left(\left(\mathbf{Y}_i^{(f)}-\mathbf{Y}_j^{(f)}\right)\cdot\left(\mathbf{Y}_i^{(f)}-\mathbf{Y}_j^{(f)}\right)^{\mathrm{T}}S_{ij}\right)$	$\sum_{c=1}^{C}\sum_{i=1}^{N_c}\left(\left(\mathbf{Y}_{c_i}^{(f)}-\bar{\mathbf{Y}}_c^{(f)}\right)\cdot\left(\mathbf{Y}_{c_i}^{(f)}-\bar{\mathbf{Y}}_c^{(f)}\right)^{\mathrm{T}}\right)$	$\mathbf{I}$
H2	$\sum_{i=1}^{N}\left(\mathbf{Y}_i^{(f)}\left(\mathbf{Y}_i^{(f)}\right)^{\mathrm{T}}D_{ij}\right)$	$\sum_{i=1}^{N}\left(\mathbf{Y}_i^{(f)}(\mathbf{Y}_i^{(f)})^{\mathrm{T}}\right)$	$\sum_{i,j=1}^{N}\left(\mathbf{Y}_i^{(f)}-\mathbf{Y}_j^{(f)}\right)\cdot\left(\mathbf{Y}_i^{(f)}-\mathbf{Y}_j^{(f)}\right)^{\mathrm{T}}S'_{ij}$	$\sum_{c=1}^{C}\left(\left(\bar{\mathbf{Y}}_c^{(f)}-\bar{\mathbf{Y}}^{(f)}\right)\cdot\left(\bar{\mathbf{Y}}_c^{(f)}-\bar{\mathbf{Y}}^{(f)}\right)^{\mathrm{T}}\right)$	$\sum_{i=1}^{N}\left(\mathbf{Y}_i^{(f)}(\mathbf{Y}_i^{(f)})^{\mathrm{T}}\right)$

Algorithm 2 Tensor Embedding Framework Utilized in Pol-SAR Land Cover Classification

1. Extract features and construct the feature cube: extract the polarimetric backscattering data and polarimetric target decomposition parameters as features to represent data points. Each pixel is described by a patch (the pixel is in the center of the patch). Construct a feature cube $\mathcal{X}_j \in \mathbb{R}^{I_1\times I_2\times I_3}$ to represent each of the pixels, in which $I_1 \times I_2$ denotes the patch size and I_3 denotes the number of original features.
2. Construct the intrinsic and penalty matrices: Choose some pixels from the data set as training samples (unsupervised without labels and supervised with labels). According to the geometrical and statistical properties of training samples to construct the intrinsic matrix, constraints, and some prior information to construct the penalty matrix.
3. Obtain the projection matrix for each order of tensors: Utilize the pairs of matrices $\mathbf{H}_1$ and $\mathbf{H}_2$, update the projection matrix according to alternative iterations, and obtain the optimal matrices $\left\{\mathbf{U}_f^*\right\}_{f=1}^{3} \in \mathbb{R}^{P_f\times I_f}$; the detailed steps are summarized in the algorithm 1.
4. Reduce the feature dimension and classification: Employ the optimal matrices $\left\{\mathbf{U}_f^*\right\}_{f=1}^{3}$ to reduce the dimensionality of samples. With those low-dimensional features, we utilize KNN and SVM classifiers to achieve the final classification result.

equals zero. Using the (6), we can get the objective function

$$\mathbf{U}_f^*|_{f=1}^M = \underset{\mathbf{U}_f}{\operatorname{argmax}} \sum_{i=1}^{N} \|\mathcal{Y}_i \times_f \mathbf{U}_f\|_F^2. \tag{14}$$

According to the mode-f unfolding $\mathcal{Y}\times_f\mathbf{U}_f \Leftrightarrow \mathbf{U}_f\mathbf{Y}^{(f)}$ to make a tractable form

$$\mathbf{U}_f^*|_{f=1}^M = \underset{\mathbf{U}_f}{\operatorname{argmax}}\ \mathrm{tr}\left(\mathbf{U}_f \sum_{i=1}^{N}\left(\mathbf{Y}_i^{(f)}(\mathbf{Y}_i^{(f)\mathrm{T}})\mathbf{U}_f^{\mathrm{T}}\right)\right). \tag{15}$$

It is due to the maximum optimization problem, then the penalty matrix is $\mathbf{H}_2 = \sum_{i=1}^{N}(\mathbf{Y}_i^{(f)}(\mathbf{Y}_i^{(f)})^{\mathrm{T}})$ in the objective function. While the constraint of TPCA is just to make the $\mathbf{U}_f^*|_{f=1}^M$ with orthogonal columns, it can be formulated as $\mathbf{U}_f\mathbf{U}_f^T = \mathbf{I}$. Then, the intrinsic matrix $\mathbf{H}_1$ is set as the infinity matrix. It does not need to use the constraint term from (11), and the maximum optimization can be directly calculated through eigenvalue decomposition $\mathbf{H}_2\alpha = \lambda\alpha$ to get the corresponding eigenvectors.

2) TNPE: TNPE as TPCA search for M optimal projection matrices $\{\mathbf{U}_f^*\}_{f=1}^M$, but the goal is to ensure projected samples maintaining the local topological structure of original manifold data. Similar to the NPE algorithm, TNPE also needs to build a similarity matrix to represent the graph structure and capture the local structure of the input samples. As for the linearization of NPE, we utilize the same method to measure the similarity between points

$$S_{ij} = \begin{cases} \exp\left(-\|\mathcal{X}_i - \mathcal{X}_j\|_{\mathrm{F}}^2\right)/t, & \mathcal{X}_j \in O_K(\mathcal{X}_i) \\ 0, & \text{otherwise} \end{cases} \tag{16}$$

in which t is the Gaussian kernel bandwidth, $\mathcal{X}_j \in O_K(\mathcal{X}_i)$ denotes that the tensor $\mathcal{X}_j$ belongs to the k-nearest neighbors of tensor $\mathcal{X}_i$. This neighborhood can also be constructed by the method of ε-neighborhood. It means one needs to measure the distance of $\mathcal{X}_i$ between the rest of the tensors and utilize a threshold value ε to limit the size of the neighborhood. If the distance between $\mathcal{X}_i$ and $\mathcal{X}_j$ is smaller than ε, it means $\mathcal{X}_j$ belongs to the neighbor of $\mathcal{X}_i$. In TNPE and the following tensorial methods, the distance utilized for measuring tensors is the Euclidean distance, which is defined in Definition 6. After the similarity matrix is constructed, each point can be represented as a linear combination of its neighbor points, and the representation coefficients are specified in the similarity matrix. Then, we can formulate the objective function to obtain a group of optimal projection matrices such that the manifold structure is preserved in low-dimensional tensor space

$$\mathbf{U}_f^*|_{f=1}^M = \underset{\mathbf{U}_f}{\operatorname{argmin}} \sum_{i=1}^{N}\left\|\mathcal{Z}_i - \sum_{j=1}^{N} S_{ij}\mathcal{Z}_j\right\|_F^2$$
$$\text{s.t.} \sum_{i=1}^{N}\|\mathcal{Z}_i\|_F^2 = c. \tag{17}$$

The constraint is to eliminate the arbitrary projection vectors in the transformation matrices, and c is a constant; here, we can set it as 1. Then, it can be reformulated from a tensorial form into a matrix form, and

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is transformed from the Frobenius norm of tensors into a tractable form of matrices. It has the same objective function as (10). Then, a pair of matrices is denoted as $\mathbf{H}_1 = \sum_{i=1}^{N} ((\mathbf{Y}_i^{(f)} - \sum_{j=1}^{N} S_{ij}\mathbf{Y}_i^{(f)})(\mathbf{Y}_i^{f} - \sum_{j=1}^{N} S_{ij}\mathbf{Y}_i^{(f)})^{\mathrm{T}})$ and $\mathbf{H}_2 = \sum_{i=1}^{N} (\mathbf{Y}_i^{(f)}(\mathbf{Y}_i^{(f)})^{\mathrm{T}})$. Each optimal transformation matrix can be obtained in an alternative iteration process. Fixing other projection matrices $U_e, e = 1, 2, \ldots, f-1, f+1, \ldots, M$, we can update $\mathbf{U}_f$ from the eigenvalue decomposition problem.

3) TLDE: This algorithm is a supervised method that is different from the above two algorithms in how to construct the similarity matrices. The target of this algorithm is to obtain better discriminant features by integrating the relation of neighbors and classes between samples. One needs to preserve the neighbor points in same class, and at the same time separate the neighbor points into different classes in the embedded tensor space. Then, the new input points can be well projected under this criterion and have better performances in the pattern classification task.

The core of TLDE is the same as for LDE. It also needs to construct a pair of within-class graph G and between-class graph G'. When constructing G, we need to connect the points $\mathcal{X}_i$ and $\mathcal{X}_j$ under the condition that they are neighbors and of the same class. On the contrary, we consider the pair $\mathcal{X}_i$ and $\mathcal{X}_j$ not in the same class, and connect them if they belong to the neighborhood of each other. The way how to build the neighborhood can select the K-nearest neighbor or ϵ-neighbor method. The graph G and G' can be identified in those similarity matrices

$$S_{ij} = \begin{cases} \exp\left(-\|\mathcal{X}_i - \mathcal{X}_j\|_{\mathrm{F}}^2\right)/t, & \begin{array}{l}\mathcal{X}_j \in O_K(\mathcal{X}_i) \\ \text{or } \mathcal{X}_i \in O_K(\mathcal{X}_j) \\ \text{and } \ l_i = l_j\end{array} \\ 0, & \text{otherwise} \end{cases} \tag{18}$$

and

$$S_{ij} = \begin{cases} \exp\left(-\|\mathcal{X}_i - \mathcal{X}_j\|_{\mathrm{F}}^2\right)/t, & \begin{array}{l}\mathcal{X}_j \in O_K(\mathcal{X}_i) \\ \text{or } \mathcal{X}_i \in O_K(\mathcal{X}_j) \\ \text{and } \ l_i \neq l_j\end{array} \\ 0, & \text{otherwise} \end{cases} \tag{19}$$

in which l_i and l_j are the labels of tensors $\mathcal{X}_i$ and $\mathcal{X}_j$. The objective function can be formulated in the Frobenius form as

$$\begin{aligned} \mathbf{U}_f^*|_{f=1}^M &= \underset{\mathbf{U}_f}{\operatorname{argmin}} \sum_{i,j=1}^{N} \|\mathcal{Z}_i - \mathcal{Z}_j\|_F^2 S_{ij} \\ &\text{s.t.} \sum_{i,j=1}^{N} \|\mathcal{Z}_i - \mathcal{Z}_j\|_F^2 S'_{ij} = c. \end{aligned} \tag{20}$$

Then, we can rewrite this form in a tractable form and replace the tensor in a matrix representation. This optimization problem can also be reformulated with the same form as in (10) and solved by eigenvalue decomposition. The pair of matrices $\mathbf{H}_1 = \sum_{i,j=1}^{N} ((\mathbf{Y}_i^{(f)} - \mathbf{Y}_j^{(f)})(\mathbf{Y}_i^{f} - \mathbf{Y}_j^{f})^{\mathrm{T}} S_{ij})$ and $\mathbf{H}_2 = \sum_{i,j=1}^{N} ((\mathbf{Y}_i^{f} - \mathbf{Y}_j^{f})(\mathbf{Y}_i^{f} - \mathbf{Y}_j^{f})^{\mathrm{T}} S'_{ij})$. This minimization problem transforms into a maximization problem to get the largest eigenvalue and the corresponding eigenvectors in eigenvalue decomposition. Each of $\mathbf{U}_f$ is updated one time in one iteration and utilized in the next iteration to update the next projection matrix.

4) TLDA: It is also a supervised method. Different from the above unsupervised methods, it seeks for a low-dimensional embedded space and gets a better representation. It seeks for an effective projection for discrimination. For TLDA, the class separability criterion is defined through two scatter tensors similar to traditional LDA. The within-tensor scatter captures the scatter of the samples around the corresponding class means. The between-tensor scatter describes the scatter of class means and the whole mean. For preserving the within-class samples, more similar and between-class samples are more distinctive, the TLDA objective function can be defined as follows:

$$\mathbf{U}_f^*|_{f=1}^M = \underset{\mathbf{U}_f}{\operatorname{argmax}} \frac{\sum_{c=1}^{C} N_c \|\bar{\mathcal{Z}}_c - \bar{\mathcal{Z}}\|_F^2}{\sum_{c=1}^{C}\sum_{i=1}^{N_c} \|\mathcal{Z}_{c_i} - \bar{\mathcal{Z}}_c\|_F^2}. \tag{21}$$

In this equation, $\mathcal{Z}_{c_i}$ denotes the ith samples belonging to class $c (c = 1, 2, \ldots, \mathrm{C})$. $\bar{\mathcal{Z}} = (1/N)\sum_{i=1}^{N} \mathcal{Z}_i$ denotes the overall mean of samples. $\bar{\mathcal{Z}}_c = (1/N_c)\sum_{i=1}^{N} \mathcal{Z}_{c_i}$ is the c class mean, in which N_c is the number of samples in the cth class. This problem can also be solved via an alternative iteration method. The two means of samples can be reformulated as $\bar{\mathcal{Z}} = \bar{\mathcal{Y}} \times_f \mathbf{U}_f$ and $\bar{\mathcal{Z}}_c = \bar{\mathcal{Y}}_c \times_f \mathbf{U}_f$ respectively, where $\bar{\mathcal{Y}} = (1/N)\sum_{i=1}^{N} \mathcal{Y}_i$ and $\bar{\mathcal{Y}}_c = (1/N_c)\sum_{i=1}^{N_c} \mathcal{Y}_{c_i}$($\mathcal{Y}_{c_i}$ denotes $\mathcal{Y}_i$ belonging to cth class). Then, each iteration needs to utilize the mode-f matrix for updating. We can also get the matrices transformed from the corresponding tensors by (6) and the mode-f unfolding $\mathcal{Y} \times_f \mathbf{U}_f \Leftrightarrow \mathbf{U}_f \mathbf{Y}^{(f)}$. Hence, the optimization of TLDA can be rewritten in the following trace form as other tensor embedding algorithms:

$$\mathbf{U}_f^*|_{f=1}^M = \underset{\mathbf{U}_f}{\operatorname{argmax}} \frac{\operatorname{tr}\left(\mathbf{U}_f \mathbf{H}_2 \mathbf{U}_f^T\right)}{\operatorname{tr}\left(\mathbf{U}_f \mathbf{H}_1 \mathbf{U}_f^T\right)}. \tag{22}$$

From this function, we can get a pair of matrices $\mathbf{H}_2 = \sum_{c=1}^{C} ((\bar{\mathbf{Y}}_c^{(f)} - \bar{\mathbf{Y}}^{(f)})(\bar{\mathbf{Y}}_c^{(f)} - \bar{\mathbf{Y}}^{(f)})^{\mathrm{T}})$ and $\mathbf{H}_1 = \sum_{c=1}^{C}\sum_{i=1}^{N_c} ((\mathbf{Y}_{c_i}^{(f)} - \bar{\mathbf{Y}}_c^{(f)})(\mathbf{Y}_{c_i}^{(f)} - \bar{\mathbf{Y}}_c^{(f)})^{\mathrm{T}})$. Then, the optimal matrix $\mathbf{U}_f^*$ is constructed by the eigenvectors corresponding to the P_f largest eigenvalues in the eigenvalue decomposition $(\mathbf{H}_2 - \gamma \mathbf{H}_1)\alpha = \lambda\alpha$. Other projection matrixes can also be obtained in each iteration.

D. Complexity Analysis

Because of the alternative optimization steps, the tensorial algorithms need to make convergence through T iterations. Then, we analyze the computational time complexity in each iteration. The complexity is mainly divided into three parts: the mode-n tensor-matrix product in (6) for each tensor samples, penalty and intrinsic matrix construction, and the eigenvalue decomposition.

The number of reduced dimensions is small enough, which is regarded as constants. For a three-order tensor, the dimension of the original tensor in each mode is I_1, I_2, and I_3. As shown in Algorithm 1, every tensorial algorithm has a

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complexity $O(\prod_{i=1}^{3} I_i)$ and $O(\sum_{i=1}^{3} I_i^3)$ in the tensor-matrix production and eigen-value decomposition step, respectively. Besides, there exist differences in constructing the pair of matrices, and those tensorial algorithms produce different time complexities. In this step, TLDE and TLPP have the same cost $O(N^2 \sum_{i=1}^{3} I_i^2)$, TPCA and TLDA have the same cost $O(N \sum_{i=1}^{3} I_i^2)$, and TNPE has $O(\sum_{i=1}^{3} (NI_i^2 + N^2 I_i))$, respectively. On the contrary, linear techniques, such as NPE, LPP, PCA, LDE, and LDA, if not considering the spatial information, have the complexity $O(I_3^3)$. But if those algorithms process the same features from a patch as tensorial techniques, it will have a higher complexity $O((\prod_{i=1}^{3} I_i)^3)$. The space complexities of those tensorial methods are $O(N \prod_{i=1}^{3} I_i)$. The linear methods have the space complexity $O(I_3^2)$ and the complexity $O((\prod_{i=1}^{3} I_i)^2)$ processing the features from the patch.

IV. Experiment and Analysis

In this section, three real PolSAR data are utilized to demonstrate the effectiveness of the proposed methods. The performances are compared with several state-of-the-art algorithms. Support vector machine (SVM) and k-nearest neighborhood (KNN) classifiers are utilized for a final PolSAR land cover classification.

A. Data Sets

The efficiency of tensorial DR algorithms in this unified framework is demonstrated on three data sets from the RADARSAT2 satellite (Canadian Space Agency). The first data set covers the region of Xianyang and the western part of Xi'an, Shaanxi province, China, in January 2010. This sensor worked in the C-band, and the data are produced with the incident angle $40° - 50°$. The image is cut from a big picture. This data set covers 512×512 pixels including three terrains: buildings, grass, and water in Fig. 3. The spatial resolution is 10×5 m.

The second data set is a well-known Flevoland data set, which covers the Flevoland region in the Netherlands, produced in April 2008. The size of the original image is 1600×2400 pixels with a resolution of 10×5 m. It has an available ground truth image for remote sensing image segmentation and land cover classification. We clip one typical region with the size of 796×769 from the original, including woodland, urban, cropland, and water in Fig. 4.

The third data set is also a C-band RADARSAT2 data set, which covers the region of the golden bridge in San Francisco. It is one of the most used regions for PolSAR land cover classification. The original image contains 1419×1895 pixels. We get the subimage with the size of 814×781. This region covers four kinds of categories, including the high-density urban, water, vegetation, and low-density urban, labeled by four colors in the ground truth image in Fig. 6.

B. Contrast Algorithms

To demonstrate the effectiveness of the proposed tensor embedding framework in PolSAR land cover classification, several state-of-the-art algorithms are selected. Linear algorithms contain the unsupervised ICA, PCA, linearization of manifold learning methods (LPP and NPE), and supervised LDA. The window size for the extracted features is set as 9×9, which will be analyzed in the following section. The neighbor number in graph embedding algorithms is set as 15. We directly use the FastICA package to test ICA algorithms. Those algorithms reduce the features in a fixed-dimension for a fair comparison. The codes of PCA, LPP, and NPE algorithms employed in comparison are from the DR toolbox of Laurens van der Maaten [58], [59].

We select unsupervised methods (TLDA, TNPE, and TLPP) and a supervised method (TLDA) in the tensorial framework for comparison. General tensor discriminant analysis (GTDA) as a typical variant of the multilinear discriminant analysis algorithm is employed to validate the tensorial LDA. TLDE is a supplement to demonstrate the effectiveness of the supervised tensorial algorithm for PolSAR data. The neighborhood size of intrinsic and penalty graphs in TLDE is also set as 15. In optimization, the tunable parameter γ for $(\mathbf{H}_2 - \gamma \mathbf{H}_1)\alpha = \lambda\alpha$ is set as 0.3. We construct the feature cube to represent each pixel in images. A tensor is of three-order $\{\mathcal{X}_i\}_{i=1}^{N} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$; I_1 and I_2 are the length and the width of the feature patch; and I_3 denotes the number of feature dimensions.

C. Experiment Preparation

Before the experiments, we need to make sufficient preparation to analyze the influence of parameters for the classification. How to extract the tensor samples and utilize those samples to construct graphs is an important issue. Those pixels are represented as feature cubes in the tensorial embedding framework. The height, width, and feature dimensions of the cube all influence the algorithm performance. When processing those data sets, we randomly select the training samples from the whole samples. The tensor samples have the advantages over the pixel samples in retaining the spatial property, and can cover more useful information for classification. Meanwhile, the number of training samples is a significant factor in learning projection matrices, which also needs to be analyzed.

We utilize two typical algorithms (GTDA and TPCA) representing a tensor embedding framework to investigate the variation of performance with changing training number and the feature patch size. It is due to those three data sets that are all produced from RADARSAT2 and have the same resolution, the size of the feature patch size varies in the range (3×3, 5×5, 7×7, 9×9, 13×13). After analyzing, we utilize the suitable patch to construct $I_1 \times I_2 \times I_3$ cubes.

Fig. 2 presents the overall classification performance with the variation of the number of training samples and the patch size. In the direction of training number, it shows that OA increases with the growing training number. As the training number is greater than 10, the OA growth will remain steady and at a high level. Therefore, some of the following experiments are conducted with the training number greater than or equal to 20. With the increase of the patch size, OA improves slowly and a balanced growth is obtained after the

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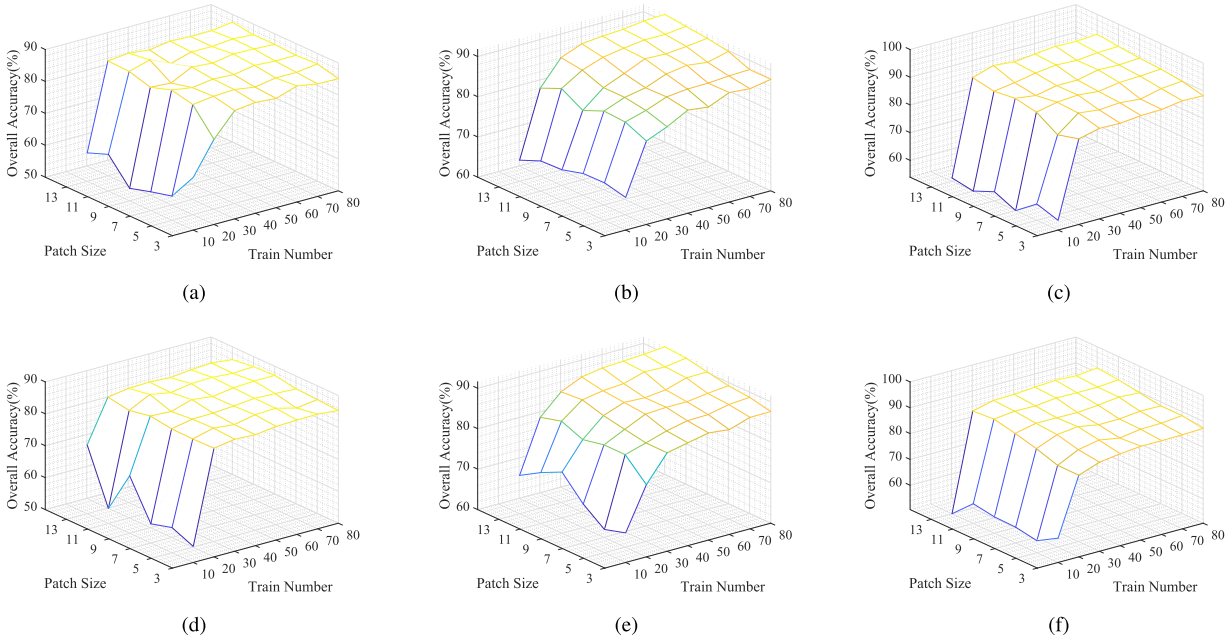

Fig. 2. OA of the KNN classifier with variation of the patch size and the train number. (a) Xi'an data set with GTDA. (b) Flevoland data set with GTDA. (c) San Francisco data set with GTDA. (d) Xi'an data set with TPCA. (e) Flevoland data set with TPCA. (f) San Francisco data set with TPCA.

9×9 size. The bigger the feature patch is, the more geographic and spatial information will be captured. Thus, a reasonably large patch can achieve a better classification result. For the three kinds of RADARSAT2 data sets, the patch sizes are all set to 9×9. This size can get a good performance enough without large calculation and only needs a less RAM requirement.

D. Experiment Results

In this section, we conduct algorithms in a tensorial embedding framework on three data sets to demonstrate the effectiveness of the extracting feature for the PolSAR land cover classification. For a fair comparison, five linear and five tensorial DR methods are utilized to process 22-D high-dimensional features. They are constructed by the original nine freedom parameters from the coherency matrix, and the coefficients of different components from four selected target decomposition mechanisms, including Freeman, VanZyl, Yamaguchi, and Krogager decomposition. Freeman and Yamaguchi decompositions are incoherent decompositions on the covariance matrix C, mainly describing the surface covered by natural vegetation and the man-made urban areas. Krogager decomposition is a coherent decomposition to resolve scattering characteristics independent of the incident angle. VanZyl decomposition is the first-considered eigenvector analysis to describe a natural terrain in a monostatic case. Those target decomposition features are redundant enough, and sensitive to different terrains and land covers. The constructed feature vectors can be combined to extract discriminant feature vectors.

For tensorial algorithms, each pixel is represented as $\mathcal{X}_i \in \mathbb{R}^{I_1 \times I_2 \times I_3}$, in which I_3 equals 22. Then, we input the same size features into linear DR techniques. As PolSAR original data can be represented by the coherency matrix, the nine freedom parameters are then directly processed by KNN and SVM, and all the DR techniques should reduce the high-dimensional features to the same dimensionality. Due to the significance of Wishart classifier in PolSAR data classification, it is regarded as a benchmark.

As discussed above, a feature patch covers 9×9 pixels. To evaluate the land cover classification performance, the training samples are selected randomly from the total samples, and we test all the methods repeatedly ten times to get the mean value to compare. The classification results are reported in three data sets with different typical land covers. Four evaluation indicators including the overall accuracy (OA), average accuracy (AA), classe-specific accuracy, and Kappa coefficients are compared. Those quantification and visualization results are shown in Figs. 3–6 and Tables III–V. In those figures, classification maps are randomly chosen from one of the repeated experiments. For solving the out-of-sample problems, we randomly choose 50 training samples to learn the projection matrix $\mathbf{P}$ for linear DR methods and a set of projection matrices $\mathbf{U}_f \in \mathbb{R}^{P_f \times I_f} (f = 1, 2, 3)$ for tensorial methods. We also use 50 samples for training KNN and SVM classifiers. In KNN, only one parameter k, the number of nearest neighbors, is set as 21. In SVM, the radial basis function (RBF) is used to train the model, and a three cross-validation is used to seek the optimal parameters. The important two parameters (kernel width and penalty coefficients) are selected from the range (0.0001 0.001 0.01 0.1 1 10 100 1000 10000). Because the parameter variation in SVM influences the classification performance too much, we only choose the KNN classification map to present the visualization results.

From those classification results, one can see that the proposed method can obtain much better classification

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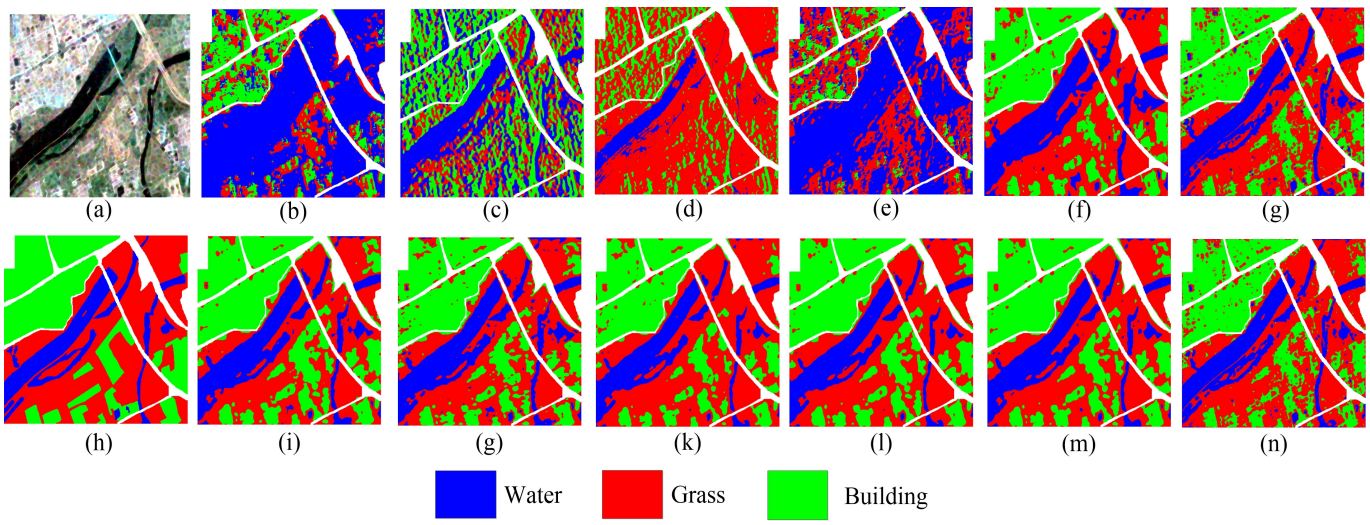

Fig. 3. Classification results obtained with the KNN classifier of different methods on RADARSAT2 Xi'an data set. (a) Pauli image. (b) ICA. (c) LDA. (d) LPP. (e) NPE. (f) PCA. (g) Original data. (h) Ground truth. (i) GTDA. (j) TLDE. (k) TLPP. (l) TNPE. (m) TPCA. (n) Wishart classifier.

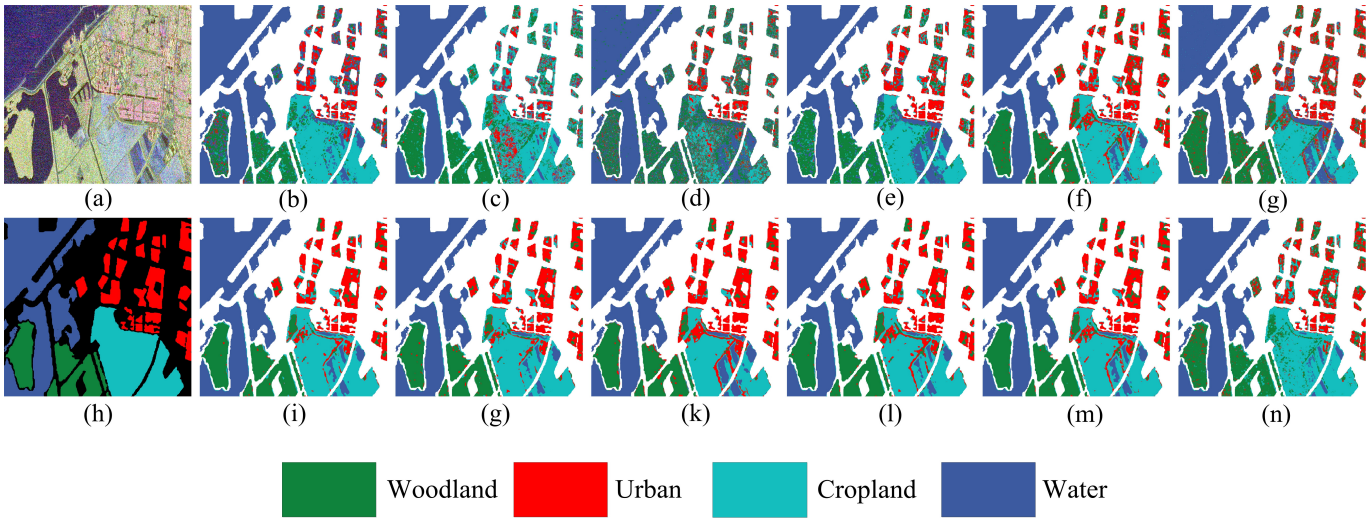

Fig. 4. Classification results obtained with the KNN classifier of different methods on the RADARSAT2 Flevoland data set. (a) Pauli image. (b) ICA. (c) LDA. (d) LPP. (e) NPE. (f) PCA.(g) Original data. (h) Ground truth. (i) GTDA. (j) TLDE. (k) TLPP. (l) TNPE. (m) TPCA. (n) Wishart classifier.

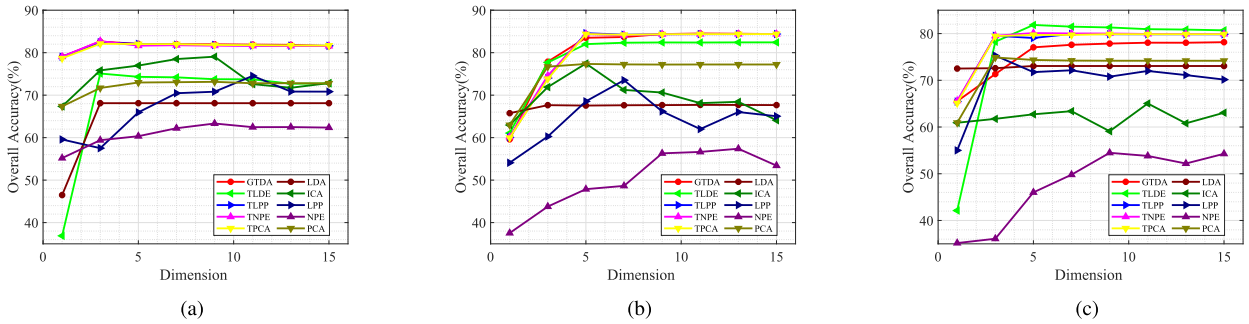

Fig. 5. OA with the dimension variation in three PolSAR data by the KNN classifier. (a) Xi'an data set. (b) Flevoland data set. (c) San Francisco data set.

performances than other state-of-the-art methods in AA, OA, Kappa coefficients, and classification maps. In Tables III–V, all the tensorial embedding methods achieve the best OA and AA results; for example, TLDA gets the value 86.85% in terms of OA in Table III that is almost higher than the best linear DR technique PCA 1.5 percent. In Tables IV and V, we can see that almost all the best results are obtained by tensorial methods, especially for TLDA. The supervised methods get all the best indicators in RADARSAT2 San Francisco and Flevoland data sets. In those tables, we assign different

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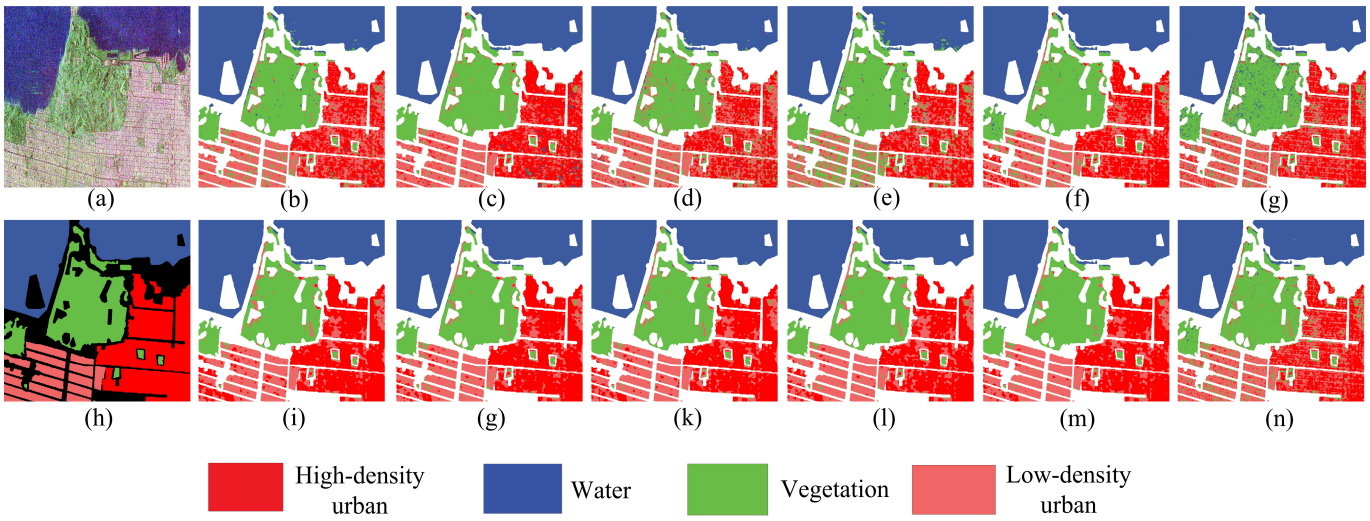

Fig. 6. Classification results obtained with the KNN classifier of different methods on the RADARSAT2 San Francisco data set. (a) Pauli image. (b) ICA. (c) LDA. (d) LPP. (e) NPE. (f) PCA. (g) Original data. (h) Ground truth. (i) GTDA. (j) TLDE. (k) TLPP. (l) TNPE. (m) TPCA. (n) Wishart classifier.

TABLE III
OA(%), AA(%), CLASS-SPECIFIC ACCURACY(%), AND KAPPA COEFFICIENT ON THE RADARSAT2 XI'AN DATA SET WITH 9-D

	Result	Original data	ICA	LDA	LPP	NPE	PCA	GTDA	TLDE	TLPP	TNPE	TPCA	Wishart
1	KNN	95.36	93.04	91.82	92.87	**95.68**	92.60	90.31	89.99	90.17	91.25	91.29	94.25
	SVM	94.42	88.35	0.03	91.52	89.70	**91.74**	89.85	83.48	88.47	87.70	87.78	
2	KNN	75.88	53.41	76.77	68.95	54.57	82.92	88.14	85.66	87.26	**88.15**	86.36	82.16
	SVM	80.89	89.70	0.14	78.70	3.39	80.60	87.69	81.81	**89.19**	88.37	88.47	
3	KNN	84.22	90.34	83.24	84.32	75.47	85.05	84.88	79.16	**85.56**	83.60	84.27	83.4999
	SVM	79.82	74.26	99.99	82.60	26.87	**80.11**	71.69	69.92	74.26	73.94	75.70	
OA	KNN	83.85	77.69	82.24	80.17	71.12	85.43	**86.85**	83.09	86.85	86.36	86.06	84.64
	SVM	82.39	81.84	49.62	82.56	28.01	82.03	80.08	76.16	81.68	81.11	**83.03**	
AA	KNN	85.15	78.93	83.94	82.05	75.24	86.86	**87.78**	84.94	87.66	87.67	87.31	86.64
	SVM	85.04	84.10	33.35	84.27	39.99	84.15	83.08	78.40	83.97	83.33	**84.26**	
Kappa	KNN	0.7228	0.6248	0.7153	0.6777	0.5380	0.7624	**0.7856**	0.7270	0.7855	0.7787	0.7732	0.7503
	SVM	0.7158	0.7095	0.3267	0.7160	0.2428	0.7101	0.6842	0.6211	0.7076	0.6975	**0.7732**	

categories of land covers with the corresponding numbers. The numbers 1–3 represent the regions of grass, building, and water in Table III, respectively. Meanwhile, the numbers 1–4 denote the regions of water, cropland, woodland, and urban in Table IV, respectively. Those four numbers also represent water, vegetation, low-density urban, and high-density urban in Table V. In Fig. 3(a) and (h), Fig. 4(a) and (h), Fig. 5(a) and (h), and Fig. 6(a) and (h), a Pauli-colored pseudoimage and a ground truth image are shown.

In the RADARSAT2 Xi'an data set, it can be seen from Fig. 3 that the methods from the tensor embedding framework all can achieve a better region uniformity compared to other linear embedding methods. In particular, in Fig. 3(i) and (m), they have shown a better performance in the grass and the building area and distinguish those two regions well. It can be seen from the results of the Wishart classifier that it cannot give a better homogeneous performance, which is also similar to the classification by the original 9-D polarimetric feature.

In the RADARSAT2 Flevoland data set, the tensorial embedding methods also obtain a better visualization performance in Fig. 4(i)–(m), which is better than the traditional linear embedding technique (first row of Fig. 4). The urban area (red) is hard to be distinguished, and we can see from the Pauli image that it mixes with some woodland. Most of the algorithms cannot classify this region well, and many urban pixels are recognized as woodland. By comparing the first row (linear embedding techniques) with the second row (tensor embedding techniques), we can find that the proposed methods provide a better spatial consistency and get a good urban region classification result.

In the RADARSAT2 San Francisco data set, we can see that almost all the methods can present a good homogeneous classification performance in regions of water and vegetation. Most of OA are higher than 90%, some algorithms even exceed 99%. Meanwhile, it should be noted that the linear embedding techniques, original data, and Wishart classifier

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TABLE IV
OA(%), AA(%), CLASS-SPECIFIC ACCURACY(%), AND KAPPA COEFFICIENT ON THE RADARSAT2 FLEVOLAND DATA SET WITH 9-D

	Result	Original data	ICA	LDA	LPP	NPE	PCA	GTDA	TLDE	TLPP	TNPE	TPCA	Wishart
1	KNN	85.86	70.25	88.51	53.10	77.56	93.32	94.51	94.76	94.81	**94.87**	94.18	89.47
	SVM	66.07	80.03	81.64	6.82	90.96	**94.25**	88.99	89.48	86.21	85.96	87.71	
2	KNN	68.01	73.64	65.47	28.70	61.32	73.06	**77.74**	75.91	73.11	75.25	76.93	81.95
	SVM	22.69	79.65	60.97	9.46	67.18	71.22	**85.39**	82.78	83.68	84.74	84.11	
3	KNN	99.43	99.19	97.75	96.68	99.64	99.05	98.88	99.02	**99.07**	98.78	98.90	99.00
	SVM	99.92	97.18	96.68	90.26	99.19	**99.44**	95.55	96.42	96.62	96.00	95.85	
4	KNN	49.48	44.47	27.34	41.22	55.46	65.35	78.27	77.17	79.51	**79.81**	79.38	54.30
	SVM	4.27	68.12	51.23	13.39	73.69	62.10	**84.67**	81.69	79.51	84.45	82.27	
OA	KNN	79.81	77.79	74.72	61.98	78.02	85.12	**88.88**	88.29	88.04	88.55	88.84	84.69
	SVM	56.75	84.52	76.59	42.03	84.87	84.33	**89.87**	89.06	88.45	89.37	89.01	
AA	KNN	82.70	71.89	69.77	54.93	73.49	75.69	**87.35**	86.72	86.62	87.18	87.35	81.18
	SVM	81.75	81.24	72.64	29.98	82.75	48.24	88.65	87.59	86.50	**87.79**	87.49	
Kappa	KNN	0.7147	0.6790	0.6437	0.4544	0.6847	0.7914	**0.8443**	0.8360	0.8326	0.8398	0.8437	0.7859
	SVM	0.3716	0.7837	0.6743	0.1026	0.7872	0.7793	**0.8589**	0.8476	0.8387	0.8518	0.8468	

TABLE V
OA(%), AA(%), CLASS-SPECIFIC ACCURACY(%), AND KAPPA COEFFICIENT ON THE RADARSAT2 SAN FRANCISCO DATA SET WITH 9-D

	Result	Original data	ICA	LDA	LPP	NPE	PCA	GTDA	TLDE	TLPP	TNPE	TPCA	Wishart
1	KNN	99.51	99.44	99.98	99.54	99.06	99.85	**99.99**	99.99	99.97	99.99	100.00	99.90
	SVM	86.18	99.91	99.94	99.58	97.41	99.99	99.98	99.98	99.99	**99.99**	99.99	
2	KNN	86.19	90.82	91.90	85.52	91.65	90.99	91.90	**92.30**	92.27	91.95	91.99	90.88
	SVM	63.48	93.57	91.90	74.93	94.06	88.57	94.16	**94.65**	94.24	94.40	94.17	
3	KNN	70.97	84.74	85.79	86.86	63.88	81.53	84.32	88.33	**89.11**	86.33	88.54	78.70
	SVM	7.36	86.07	79.81	65.02	44.00	84.81	92.76	93.13	94.21	**94.52**	93.53	
4	KNN	54.34	56.84	71.46	48.49	60.29	70.48	75.83	76.67	77.50	**79.42**	78.03	55.82
	SVM	2.83	78.40	76.63	5.08	58.06	64.10	85.17	86.92	86.32	**87.03**	86.34	
OA	KNN	79.01	82.85	87.87	80.90	81.82	86.33	89.44	**90.76**	90.33	90.35	90.21	82.54
	SVM	85.30	90.53	88.83	62.53	78.52	87.07	93.48	94.14	94.02	**94.30**	93.92	
AA	KNN	77.14	82.70	87.16	80.59	78.09	77.01	88.15	**89.80**	89.42	89.42	89.51	81.32
	SVM	82.91	89.49	87.07	61.15	73.38	86.58	93.02	93.67	93.69	**93.99**	93.51	
Kappa	KNN	0.7145	0.7687	0.8345	0.7426	0.7513	0.8141	0.8556	**0.8736**	0.8679	0.8681	0.8663	0.7637
	SVM	0.7986	0.8703	0.8469	0.4899	0.7057	0.7984	0.9106	0.9196	0.9179	**0.9217**	0.9166	

give bad results in the low-density urban and the high-density urban areas. OA for some of them are lower than 80%. This is due to the high similarity in those urban regions. It is even hard to directly classify them from the original features. From the observation of Fig. 6, we can arrive at the conclusion that tensorial embedding methods achieve the best visualization performance with the KNN classifier, especially in the urban areas.

E. Experiment Analysis

1) Dimensionality Analysis: We randomly choose 20 training samples from each category to learn the series of projection matrix $\mathbf{U}_f$. KNN results of three PolSAR data sets are shown in Fig. 5, as the dimensionality changes from 1 to 15 with the step length of 2. Fig. 5 shows that all the tensor embedding DR methods have a higher classification performance than linear techniques. In particular, the dimensionality is higher than 5 and all the tensor embedding can reach the best OA. It means the tensor method has low sensitivity to the training sample. After the methods retained a stable performance, all the proposed methods can get a better OA.

2) Convergence Analysis: In order to analyze the convergence of tensor embedding methods, we also choose GTDA and TPCA as examples to represent the methods unified in the tensor framework. The error between two iterations after

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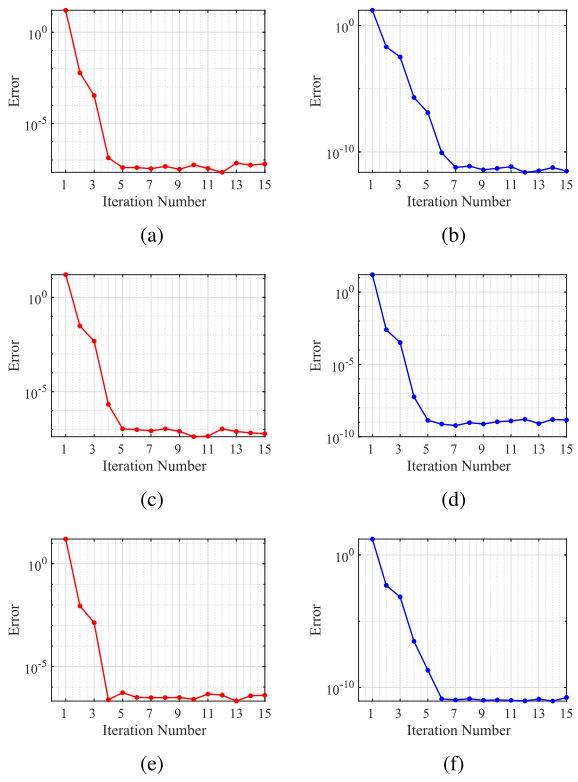

Fig. 7. Convergence of GLDA and TPCA in PolSAR data. The vertical axis indicates the error of projection matrices $\{\mathbf{U}_f\}_{f=1}^{3}$ in two iterations. (a) Xi'an data set with GTDA. (b) Xi'an data set with TPCA. (c) Flevoland data set with GTDA. (d) Flevoland data set with TPCA. (e) San Francisco data set with GTDA. (f) San Francisco data set with TPCA.

taking the logarithm is employed on three RADARSAT2 data sets. In Fig. 7, we can see that the residual error reduces along with the increase of the number of iterations. For the GTDA method, it converges after three iterations in Fig. 7, and the error is almost 0, which means the tensorial methods can converge in a few iterations. Meanwhile, TPCA does not have such a quick convergence speed, but can also get stable convergency after seven iterations, which shows that the supervised method can achieve a quick convergence. According to convergence analysis, the number of iterations is set to 10. This makes sure that all the tensorial methods can achieve a stable result.

V. Conclusion

In this article, we proposed a unified tensor embedding framework transforming different typical linear DR techniques into a tensorial form. In this framework, the tensorial representation highly improved the feature extraction ability for PolSAR data and provided the distinctive features for the subsequent PolSAR land cover classification. According to the geometrical and statistical properties of training samples, we can construct the intrinsic matrix and the prior constraint to construct the penalty matrix. With those pairs of matrices, the obtained projection matrices can effectively solve the out-of-sample problems. Those tensor methods combine the spatial information around the pixels, which can comprehensively describe the data characteristics in a three-order tensor. Those tensorial methods present superiority and efficiency in visualization and quantification performances compared to other linear DR methods. In the future, we will focus on studying the tensorial properties of multi-modal features in PolSAR data processing, exploiting the coherence of multi-modal space, and considering how to utilize a more suitable tensor distance to construct the similarity matrix or graphs.

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Bo Ren (M'18) received the B.S. degree in telecommunications engineering from Northwest University, Xi'an, China, in 2011, and the Ph.D. degree in circuits and systems from Xidian University, Xi'an, in 2017.

Since 2018, he has been an Assistant Professor with the Key Laboratory of Intelligent Perception and Image Understanding of the Ministry of Education of China, Xidian University. He is currently a Visiting Scholar with the Grenoble Images Parole Signal Automatique (GIPSA-Lab), Grenoble, France. His research interests include synthetic aperture radar image interpretation and understanding and sparse representation and machine learning in remote sensing images.

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Biao Hou (M'07) received the B.S. and M.S. degrees in mathematics from Northwest University, Xi'an, China, in 1996 and 1999, respectively, and the Ph.D. degree in circuits and systems from Xidian University, Xi'an, in 2003.

Since 2003, he has been with the Key Laboratory of Intelligent Perception and Image Understanding of the Ministry of Education, Xidian University, where he is currently a Professor. His research interests include compressive sensing and synthetic aperture radar image interpretation.

Jocelyn Chanussot (M'04–SM'04–F'12) received the M.Sc. degree in electrical engineering from the Grenoble Institute of Technology (Grenoble INP), Grenoble, France, in 1995, and the Ph.D. degree from the Université de Savoie, Annecy, France, in 1998.

In 1999, he was with the Geography Imagery Perception Laboratory, Delegation Generale de l'Armement (DGA—French National Defense Department). From 2015 to 2017, he was a Visiting Professor with the University of California at Los Angeles (UCLA), Los Angeles, CA, USA. Since 1999, he has been with the Grenoble INP, where he is currently a Professor of signal and image processing. He has been a Visiting Scholar with Stanford University, Stanford, CA, USA, the KTH Royal Institute of Technology, Stockholm, Sweden, and the National University of Singapore, Singapore. Since 2013, he has been an Adjunct Professor with the University of Iceland, Reykjavík, Iceland. He is currently conducting his research with the Grenoble Images Speech Signals and Automatics Laboratory (GIPSA-Lab). His research interests include image analysis, multicomponent image processing, nonlinear filtering, data fusion, and machine learning in remote sensing.

Dr. Chanussot was a member of the Machine Learning for Signal Processing Technical Committee at the IEEE Signal Processing Society from 2006 to 2008. He is a member of the Institut Universitaire de France from 2012 to 2017, a 2018 Highly Cited Researcher (Clarivate Analytics), and a member of the IEEE Geoscience and Remote Sensing Society AdCom. He was a co-recipient of the NORSIG 2006 Best Student Paper Award, the IEEE GRSS 2011 and 2015 Symposium Best Paper Award, the IEEE GRSS 2012 Transactions Prize Paper Award, and the IEEE GRSS 2013 Highest Impact Paper Award. He is the Founding President of the IEEE Geoscience and Remote Sensing French Chapter from 2007 to 2010 which received the 2010 IEEE GRS-S Chapter Excellence Award. He was the General Chair of the first IEEE GRSS Workshop on Hyperspectral Image and Signal Processing and Evolution in Remote sensing (WHISPERS). He was the Co-Chair and the Chair of the GRS Data Fusion Technical Committee from 2005 to 2008 and from 2009 to 2011, respectively, and the Program Chair of the IEEE International Workshop on Machine Learning for Signal Processing in 2009. He was the Editor-in-Chief of the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING from 2011 to 2015. He was the Guest Editor of the PROCEEDINGS OF THE IEEE in 2013 and the *IEEE Signal Processing Magazine* in 2014. He is an Associate Editor of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING and the IEEE TRANSACTIONS ON IMAGE PROCESSING.

Licheng Jiao (SM'89–F'17) was born in Shaanxi, China, in October, 1959. He received the B.S. degree from Shanghai Jiaotong University, Shanghai, China, in 1982, and the M.S. and Ph.D. degrees from Xi'an Jiaotong University, Xi'an, China, in 1984 and 1990, respectively.

From 1984 to 1986, he was an Assistant Professor with the Civil Aviation Institute of China, Tianjin, China. From 1990 to 1991, he was a Post-Doctoral Fellow with the Key Laboratory for Radar Signal Processing, Xidian University, Xi'an, where he is currently the Director of the Key Laboratory of Intelligent Perception and Image Understanding of Ministry of Education of China. His research interests include signal and image processing, nonlinear circuits and systems theory, learning theory and algorithms, optimization problems, wavelet theory, and machine learning.

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